

Import des librairies

```
import pandas as pd
import numpy as np
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import (
    accuracy_score,
    f1_score,
    precision_score,
    recall_score, confusion_matrix,
    ConfusionMatrixDisplay,
    classification_report,
    roc_curve,
    RocCurveDisplay,
    PrecisionRecallDisplay
)
from sklearn.model_selection import GridSearchCV, RandomizedSearchCV
from sklearn.svm import LinearSVC

X_train = pd.read_csv('X_train_clean.csv')
X_test = pd.read_csv('X_test_clean.csv')
y_train = pd.read_csv('y_train.csv')
y_test = pd.read_csv('y_test.csv')

print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)

(242, 17)
(61, 17)
(242, 1)
(61, 1)
```

Hyperparameters KNN

```
n_neighbors = 5
weights = 'uniform'
algorithm = 'auto'
leaf_size = 30
p = 2
metric = 'minkowski'

model_knn = KNeighborsClassifier(
    n_neighbors=n_neighbors,
    weights=weights,
    algorithm=algorithm,
    leaf_size=leaf_size,
    p=p,
```

```

        metric=metric
    )

model_knn
y_train.values.ravel()

model_knn.fit(X_train, y_train.values.ravel())
KNeighborsClassifier()

y_pred = model_knn.predict(X_test.values)

C:\Users\anasa\AppData\Roaming\jupyterlab-desktop\jlab_server\Lib\
site-packages\sklearn\base.py:493: UserWarning: X does not have valid
feature names, but KNeighborsClassifier was fitted with feature names
    warnings.warn(

# comparer les valeurs de prédiction de X_test avec y_test
y_test

   target
0        0
1        1
2        0
3        0
4        0
..      ...
56       1
57       0
58       1
59       0
60       1

[61 rows x 1 columns]

```

Etudier les metrics

```

# Exactitude : Accuracy
acc_score = accuracy_score(y_pred, y_test)
print(acc_score)

0.8360655737704918

```

GridSearchCV (Recherche du meilleur modèle)

```

# Recherche des meilleurs paramètres
hyperparametres = {
    'n_neighbors' : list(range(3, 20)),

```

```

    'weights' : ['uniform', 'distance'],
    'algorithm' : ['ball_tree', 'kd_tree'],
    'p' : list(range(2, 5))
}

gscv_knn = GridSearchCV(
    estimator=model_knn,
    param_grid=hyperparametres,
    cv=5,
    scoring= 'f1',
    verbose= 4
)

# lancer le modèle
gscv_knn.fit(X_train, y_train.values.ravel())

Fitting 5 folds for each of 204 candidates, totalling 1020 fits
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[CV 5/5] END algorithm=ball_tree, n_neighbors=15, p=4, weights=uniform;; score=0.837 total time= 0.0s

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[CV 2/5] END algorithm=ball_tree, n_neighbors=18, p=4, weights=distance;; score=0.870 total time= 0.0s
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```

```
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score=0.818 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=3, p=2, weights=distance;;
score=0.773 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=uniform;;
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=uniform;;
score=0.739 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=uniform;;
score=0.857 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=uniform;;
score=0.826 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=uniform;;
score=0.773 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=distance;;
score=0.682 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=distance;;
score=0.694 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=distance;;
score=0.800 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=distance;;
score=0.800 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=3, p=3, weights=distance;;
score=0.773 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=uniform;;
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=uniform;;
score=0.739 total time= 0.0s
```



```
[CV 3/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=uniform;;
score=0.850 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=uniform;;
score=0.826 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=uniform;;
score=0.773 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=distance;;
score=0.682 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=distance;;
score=0.694 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=distance;;
score=0.821 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=distance;;
score=0.800 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=3, p=4, weights=distance;;
score=0.773 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=uniform;;
score=0.750 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=uniform;;
score=0.744 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=uniform;;
score=0.757 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=uniform;;
score=0.810 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=uniform;;
score=0.732 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=distance;;
score=0.711 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=distance;;
score=0.739 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=distance;;
score=0.737 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=distance;;
score=0.844 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=4, p=2, weights=distance;;
score=0.773 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=uniform;;
score=0.732 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=uniform;;
score=0.744 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=uniform;;
score=0.757 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=uniform;;
score=0.791 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=uniform;;
score=0.732 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=distance;;
score=0.711 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=distance;;
```

```
score=0.696 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=distance;,
score=0.737 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=distance;,
score=0.826 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=4, p=3, weights=distance;,
score=0.773 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=uniform;,
score=0.732 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=uniform;,
score=0.744 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=uniform;,
score=0.757 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=uniform;,
score=0.791 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=uniform;,
score=0.732 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=distance;,
score=0.711 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=distance;,
score=0.723 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=distance;,
score=0.737 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=distance;,
score=0.826 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=4, p=4, weights=distance;,
score=0.773 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=uniform;,
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=uniform;,
score=0.800 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=uniform;,
score=0.821 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=uniform;,
score=0.851 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=uniform;,
score=0.800 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=distance;,
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=distance;,
score=0.739 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=distance;,
score=0.821 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=distance;,
score=0.851 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=5, p=2, weights=distance;,
score=0.800 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=uniform;,
score=0.756 total time= 0.0s
```

```
[CV 2/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=uniform;;
score=0.800 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=uniform;;
score=0.821 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=uniform;;
score=0.851 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=uniform;;
score=0.800 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=distance;;
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=distance;;
score=0.739 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=distance;;
score=0.821 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=distance;;
score=0.826 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=5, p=3, weights=distance;;
score=0.783 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=uniform;;
score=0.756 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=uniform;;
score=0.800 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=uniform;;
score=0.821 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=uniform;;
score=0.851 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=uniform;;
score=0.800 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=distance;;
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=distance;;
score=0.739 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=distance;;
score=0.821 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=distance;;
score=0.826 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=5, p=4, weights=distance;;
score=0.783 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=uniform;;
score=0.650 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=uniform;;
score=0.800 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=uniform;;
score=0.789 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=uniform;;
score=0.791 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=uniform;;
score=0.810 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=distance;;
```

```
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=distance;,
score=0.773 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=distance;,
score=0.750 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=distance;,
score=0.870 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=6, p=2, weights=distance;,
score=0.791 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=uniform;,
score=0.650 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=uniform;,
score=0.800 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=uniform;,
score=0.789 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=uniform;,
score=0.791 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=uniform;,
score=0.810 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=distance;,
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=distance;,
score=0.756 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=distance;,
score=0.750 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=distance;,
score=0.870 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=6, p=3, weights=distance;,
score=0.791 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=uniform;,
score=0.632 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=uniform;,
score=0.800 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=uniform;,
score=0.789 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=uniform;,
score=0.791 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=uniform;,
score=0.810 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=distance;,
score=0.682 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=distance;,
score=0.756 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=distance;,
score=0.769 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=distance;,
score=0.870 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=6, p=4, weights=distance;,
score=0.791 total time= 0.0s
```

```
[CV 1/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=uniform;;
score=0.739 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=uniform;;
score=0.826 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=uniform;;
score=0.750 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=uniform;;
score=0.818 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=uniform;;
score=0.837 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=distance;;
score=0.756 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=distance;;
score=0.783 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=distance;;
score=0.769 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=distance;;
score=0.844 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=7, p=2, weights=distance;;
score=0.837 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=uniform;;
score=0.739 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=uniform;;
score=0.826 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=uniform;;
score=0.780 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=uniform;;
score=0.818 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=uniform;;
score=0.837 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=distance;;
score=0.756 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=distance;;
score=0.783 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=distance;;
score=0.800 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=distance;;
score=0.844 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=7, p=3, weights=distance;;
score=0.837 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=uniform;;
score=0.739 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=uniform;;
score=0.826 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=uniform;;
score=0.780 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=uniform;;
score=0.844 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=uniform;;
```

```
score=0.837 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=distance;,
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=distance;,
score=0.783 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=distance;,
score=0.800 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=distance;,
score=0.844 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=7, p=4, weights=distance;,
score=0.837 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=uniform;,
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=uniform;,
score=0.791 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=uniform;,
score=0.769 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=uniform;,
score=0.810 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=uniform;,
score=0.780 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=distance;,
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=distance;,
score=0.756 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=distance;,
score=0.769 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=distance;,
score=0.844 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=8, p=2, weights=distance;,
score=0.780 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=uniform;,
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=uniform;,
score=0.773 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=uniform;,
score=0.750 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=uniform;,
score=0.810 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=uniform;,
score=0.780 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=distance;,
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=distance;,
score=0.739 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=distance;,
score=0.769 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=distance;,
score=0.870 total time= 0.0s
```

```
[CV 5/5] END algorithm=kd_tree, n_neighbors=8, p=3, weights=distance;;
score=0.780 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=uniform;;
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=uniform;;
score=0.773 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=uniform;;
score=0.800 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=uniform;;
score=0.810 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=uniform;;
score=0.750 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=distance;;
score=0.744 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=distance;;
score=0.739 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=distance;;
score=0.800 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=distance;;
score=0.870 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=8, p=4, weights=distance;;
score=0.762 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=uniform;;
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=uniform;;
score=0.809 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=uniform;;
score=0.780 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=uniform;;
score=0.818 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=uniform;;
score=0.810 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=distance;;
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=distance;;
score=0.783 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=distance;;
score=0.800 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=distance;;
score=0.844 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=9, p=2, weights=distance;;
score=0.810 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=9, p=3, weights=uniform;;
score=0.711 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=9, p=3, weights=uniform;;
score=0.809 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=9, p=3, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=9, p=3, weights=uniform;;
```

```
score=0.837 total time= 0.0s
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score=0.810 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=9, p=3, weights=distance;;
score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=9, p=3, weights=distance;;
score=0.783 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=9, p=3, weights=distance;;
score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=9, p=3, weights=distance;;
score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=9, p=3, weights=distance;;
score=0.810 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=uniform;;
score=0.711 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=uniform;;
score=0.809 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=uniform;;
score=0.837 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=uniform;;
score=0.810 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=distance;;
score=0.698 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=distance;;
score=0.783 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=distance;;
score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=distance;;
score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=9, p=4, weights=distance;;
score=0.810 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=10, p=2, weights=uniform;;
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=10, p=2, weights=uniform;;
score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=10, p=2, weights=uniform;;
score=0.750 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=10, p=2, weights=uniform;;
score=0.857 total time= 0.0s
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score=0.780 total time= 0.0s
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weights=distance;; score=0.698 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=10, p=2,
weights=distance;; score=0.783 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=10, p=2,
weights=distance;; score=0.769 total time= 0.0s
```



```
[CV 4/5] END algorithm=kd_tree, n_neighbors=10, p=2,
weights=distance;; score=0.818 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=10, p=2,
weights=distance;; score=0.762 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=10, p=3, weights=uniform;;
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=10, p=3, weights=uniform;;
score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=10, p=3, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=10, p=3, weights=uniform;;
score=0.829 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=10, p=3, weights=uniform;;
score=0.750 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=10, p=3,
weights=distance;; score=0.698 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=10, p=3,
weights=distance;; score=0.783 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=10, p=3,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=10, p=3,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=10, p=3,
weights=distance;; score=0.762 total time= 0.0s
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score=0.698 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=10, p=4, weights=uniform;;
score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=10, p=4, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=10, p=4, weights=uniform;;
score=0.829 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=10, p=4, weights=uniform;;
score=0.750 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=10, p=4,
weights=distance;; score=0.698 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=10, p=4,
weights=distance;; score=0.783 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=10, p=4,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=10, p=4,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=10, p=4,
weights=distance;; score=0.762 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=11, p=2, weights=uniform;;
score=0.698 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=11, p=2, weights=uniform;;
score=0.844 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=11, p=2, weights=uniform;;
```

```
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=11, p=2, weights=uniform;,
score=0.884 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=11, p=2, weights=uniform;,
score=0.727 total time= 0.0s
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weights=distance;, score=0.727 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=11, p=2,
weights=distance;, score=0.800 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=11, p=2,
weights=distance;, score=0.829 total time= 0.0s
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weights=distance;, score=0.884 total time= 0.0s
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weights=distance;, score=0.727 total time= 0.0s
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score=0.844 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=11, p=3, weights=uniform;,
score=0.810 total time= 0.0s
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score=0.884 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=11, p=3, weights=uniform;,
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[CV 1/5] END algorithm=kd_tree, n_neighbors=11, p=3,
weights=distance;, score=0.711 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=11, p=3,
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[CV 3/5] END algorithm=kd_tree, n_neighbors=11, p=3,
weights=distance;, score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=11, p=3,
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[CV 5/5] END algorithm=kd_tree, n_neighbors=11, p=3,
weights=distance;, score=0.744 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=11, p=4, weights=uniform;,
score=0.682 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=11, p=4, weights=uniform;,
score=0.844 total time= 0.0s
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score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=11, p=4, weights=uniform;,
score=0.884 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=11, p=4, weights=uniform;,
score=0.744 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=11, p=4,
weights=distance;, score=0.682 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=11, p=4,
weights=distance;, score=0.800 total time= 0.0s
```

```
[CV 3/5] END algorithm=kd_tree, n_neighbors=11, p=4,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=11, p=4,
weights=distance;; score=0.884 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=11, p=4,
weights=distance;; score=0.744 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=12, p=2, weights=uniform;;
score=0.667 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=12, p=2, weights=uniform;;
score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=12, p=2, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=12, p=2, weights=uniform;;
score=0.857 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=12, p=2, weights=uniform;;
score=0.714 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=12, p=2,
weights=distance;; score=0.698 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=12, p=2,
weights=distance;; score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=12, p=2,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=12, p=2,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=12, p=2,
weights=distance;; score=0.744 total time= 0.0s
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score=0.651 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=12, p=3, weights=uniform;;
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[CV 4/5] END algorithm=kd_tree, n_neighbors=12, p=3, weights=uniform;;
score=0.857 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=12, p=3, weights=uniform;;
score=0.714 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=12, p=3,
weights=distance;; score=0.682 total time= 0.0s
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weights=distance;; score=0.818 total time= 0.0s
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weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=12, p=3,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=12, p=3,
weights=distance;; score=0.744 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=12, p=4, weights=uniform;;
score=0.651 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=12, p=4, weights=uniform;;
score=0.818 total time= 0.0s
```

```
[CV 3/5] END algorithm=kd_tree, n_neighbors=12, p=4, weights=uniform;,
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=12, p=4, weights=uniform;,
score=0.857 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=12, p=4, weights=uniform;,
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weights=distance;, score=0.682 total time= 0.0s
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score=0.714 total time= 0.0s
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[CV 2/5] END algorithm=kd_tree, n_neighbors=13, p=2,
weights=distance;, score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=13, p=2,
weights=distance;, score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=13, p=2,
weights=distance;, score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=13, p=2,
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[CV 2/5] END algorithm=kd_tree, n_neighbors=13, p=3, weights=uniform;,
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score=0.810 total time= 0.0s
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score=0.837 total time= 0.0s
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score=0.714 total time= 0.0s
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weights=distance;, score=0.651 total time= 0.0s
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```

```
weights=distance;; score=0.800 total time= 0.0s
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weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=13, p=3,
weights=distance;; score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=13, p=3,
weights=distance;; score=0.744 total time= 0.0s
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score=0.837 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=13, p=4, weights=uniform;;
score=0.714 total time= 0.0s
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weights=distance;; score=0.651 total time= 0.0s
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weights=distance;; score=0.744 total time= 0.0s
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score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=14, p=2, weights=uniform;;
score=0.857 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=14, p=2, weights=uniform;;
score=0.714 total time= 0.0s
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weights=distance;; score=0.683 total time= 0.0s
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weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=14, p=2,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=14, p=2,
weights=distance;; score=0.744 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=14, p=3, weights=uniform;;
score=0.683 total time= 0.0s
```

```
[CV 2/5] END algorithm=kd_tree, n_neighbors=14, p=3, weights=uniform;,
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[CV 5/5] END algorithm=kd_tree, n_neighbors=14, p=3, weights=uniform;,
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weights=distance;, score=0.683 total time= 0.0s
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[CV 5/5] END algorithm=kd_tree, n_neighbors=14, p=3,
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weights=distance;, score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=14, p=4,
weights=distance;, score=0.744 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=15, p=2, weights=uniform;,
score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=15, p=2, weights=uniform;,
score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=15, p=2, weights=uniform;,
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=15, p=2, weights=uniform;,
score=0.837 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=15, p=2, weights=uniform;,
score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=15, p=2,
```

```
weights=distance;; score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=15, p=2,
weights=distance;; score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=15, p=2,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=15, p=2,
weights=distance;; score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=15, p=2,
weights=distance;; score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=15, p=3, weights=uniform;;
score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=15, p=3, weights=uniform;;
score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=15, p=3, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=15, p=3, weights=uniform;;
score=0.837 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=15, p=3, weights=uniform;;
score=0.714 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=15, p=3,
weights=distance;; score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=15, p=3,
weights=distance;; score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=15, p=3,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=15, p=3,
weights=distance;; score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=15, p=3,
weights=distance;; score=0.714 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=15, p=4, weights=uniform;;
score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=15, p=4, weights=uniform;;
score=0.844 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=15, p=4, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=15, p=4, weights=uniform;;
score=0.837 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=15, p=4, weights=uniform;;
score=0.714 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=15, p=4,
weights=distance;; score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=15, p=4,
weights=distance;; score=0.844 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=15, p=4,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=15, p=4,
weights=distance;; score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=15, p=4,
weights=distance;; score=0.714 total time= 0.0s
```

```
[CV 1/5] END algorithm=kd_tree, n_neighbors=16, p=2, weights=uniform;;
score=0.650 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=16, p=2, weights=uniform;;
score=0.837 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=16, p=2, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=16, p=2, weights=uniform;;
score=0.857 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=16, p=2, weights=uniform;;
score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=16, p=2,
weights=distance;; score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=16, p=2,
weights=distance;; score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=16, p=2,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=16, p=2,
weights=distance;; score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=16, p=2,
weights=distance;; score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=16, p=3, weights=uniform;;
score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=16, p=3, weights=uniform;;
score=0.837 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=16, p=3, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=16, p=3, weights=uniform;;
score=0.857 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=16, p=3, weights=uniform;;
score=0.714 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=16, p=3,
weights=distance;; score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=16, p=3,
weights=distance;; score=0.818 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=16, p=3,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=16, p=3,
weights=distance;; score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=16, p=3,
weights=distance;; score=0.714 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=16, p=4, weights=uniform;;
score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=16, p=4, weights=uniform;;
score=0.864 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=16, p=4, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=16, p=4, weights=uniform;;
score=0.837 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=16, p=4, weights=uniform;;
```



```
score=0.714 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=16, p=4,
weights=distance;; score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=16, p=4,
weights=distance;; score=0.844 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=16, p=4,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=16, p=4,
weights=distance;; score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=16, p=4,
weights=distance;; score=0.714 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=17, p=2, weights=uniform;;
score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=17, p=2, weights=uniform;;
score=0.864 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=17, p=2, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=17, p=2, weights=uniform;;
score=0.837 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=17, p=2, weights=uniform;;
score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=17, p=2,
weights=distance;; score=0.650 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=17, p=2,
weights=distance;; score=0.844 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=17, p=2,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=17, p=2,
weights=distance;; score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=17, p=2,
weights=distance;; score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=17, p=3, weights=uniform;;
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=17, p=3, weights=uniform;;
score=0.889 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=17, p=3, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=17, p=3, weights=uniform;;
score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=17, p=3, weights=uniform;;
score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=17, p=3,
weights=distance;; score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=17, p=3,
weights=distance;; score=0.870 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=17, p=3,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=17, p=3,
weights=distance;; score=0.889 total time= 0.0s
```

```
[CV 5/5] END algorithm=kd_tree, n_neighbors=17, p=3,
weights=distance;; score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=17, p=4, weights=uniform;;
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=17, p=4, weights=uniform;;
score=0.913 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=17, p=4, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=17, p=4, weights=uniform;;
score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=17, p=4, weights=uniform;;
score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=17, p=4,
weights=distance;; score=0.700 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=17, p=4,
weights=distance;; score=0.870 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=17, p=4,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=17, p=4,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=17, p=4,
weights=distance;; score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=18, p=2, weights=uniform;;
score=0.650 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=18, p=2, weights=uniform;;
score=0.864 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=18, p=2, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=18, p=2, weights=uniform;;
score=0.810 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=18, p=2, weights=uniform;;
score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=18, p=2,
weights=distance;; score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=18, p=2,
weights=distance;; score=0.826 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=18, p=2,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=18, p=2,
weights=distance;; score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=18, p=2,
weights=distance;; score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=18, p=3, weights=uniform;;
score=0.650 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=18, p=3, weights=uniform;;
score=0.889 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=18, p=3, weights=uniform;;
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=18, p=3, weights=uniform;;
```

```
score=0.837 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=18, p=3, weights=uniform;,
score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=18, p=3,
weights=distance;; score=0.683 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=18, p=3,
weights=distance;; score=0.851 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=18, p=3,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=18, p=3,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=18, p=3,
weights=distance;; score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=18, p=4, weights=uniform;,
score=0.650 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=18, p=4, weights=uniform;,
score=0.889 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=18, p=4, weights=uniform;,
score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=18, p=4, weights=uniform;,
score=0.837 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=18, p=4, weights=uniform;,
score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=18, p=4,
weights=distance;; score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=18, p=4,
weights=distance;; score=0.870 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=18, p=4,
weights=distance;; score=0.829 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=18, p=4,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=18, p=4,
weights=distance;; score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=19, p=2, weights=uniform;,
score=0.714 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=19, p=2, weights=uniform;,
score=0.870 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=19, p=2, weights=uniform;,
score=0.791 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=19, p=2, weights=uniform;,
score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=19, p=2, weights=uniform;,
score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=19, p=2,
weights=distance;; score=0.700 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=19, p=2,
weights=distance;; score=0.851 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=19, p=2,
weights=distance;; score=0.810 total time= 0.0s
```

```

[CV 4/5] END algorithm=kd_tree, n_neighbors=19, p=2,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=19, p=2,
weights=distance;; score=0.698 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=19, p=3, weights=uniform;;
score=0.650 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=19, p=3, weights=uniform;;
score=0.870 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=19, p=3, weights=uniform;;
score=0.791 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=19, p=3, weights=uniform;;
score=0.889 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=19, p=3, weights=uniform;;
score=0.682 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=19, p=3,
weights=distance;; score=0.667 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=19, p=3,
weights=distance;; score=0.851 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=19, p=3,
weights=distance;; score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=19, p=3,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=19, p=3,
weights=distance;; score=0.682 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=19, p=4, weights=uniform;;
score=0.650 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=19, p=4, weights=uniform;;
score=0.870 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=19, p=4, weights=uniform;;
score=0.791 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=19, p=4, weights=uniform;;
score=0.913 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=19, p=4, weights=uniform;;
score=0.711 total time= 0.0s
[CV 1/5] END algorithm=kd_tree, n_neighbors=19, p=4,
weights=distance;; score=0.667 total time= 0.0s
[CV 2/5] END algorithm=kd_tree, n_neighbors=19, p=4,
weights=distance;; score=0.851 total time= 0.0s
[CV 3/5] END algorithm=kd_tree, n_neighbors=19, p=4,
weights=distance;; score=0.810 total time= 0.0s
[CV 4/5] END algorithm=kd_tree, n_neighbors=19, p=4,
weights=distance;; score=0.864 total time= 0.0s
[CV 5/5] END algorithm=kd_tree, n_neighbors=19, p=4,
weights=distance;; score=0.711 total time= 0.0s

```

```

GridSearchCV(cv=5, estimator=KNeighborsClassifier(),
             param_grid={'algorithm': ['ball_tree', 'kd_tree'],
                         'n_neighbors': [3, 4, 5, 6, 7, 8, 9, 10, 11,
12, 13,
14, 15, 16, 17, 18, 19]},

```

```

        'p': [2, 3, 4], 'weights': ['uniform',
'distance']},
        scoring='f1', verbose=4)

# Chercher les meilleurs hyperparamètres correspondants
best_knn = gscv_knn.best_estimator_
print(best_knn)

KNeighborsClassifier(algorithm='ball_tree', n_neighbors=7, p=4)

best_knn_hyperparameters = gscv_knn.best_params_
print(best_knn_hyperparameters)

{'algorithm': 'ball_tree', 'n_neighbors': 7, 'p': 4, 'weights':
'uniform'}

# Il y a plus d'exemple dans le train que dans le test
y_pred = best_knn.predict(X_test)

y_pred.shape
(61,)

y_test.shape
(61, 1)

print("Accuracy :", accuracy_score(y_pred, y_test.values))
print("Precision :", precision_score(y_pred, y_test.values))
print("Recall :", recall_score(y_pred, y_test.values))
print("F1 :", f1_score(y_pred, y_test.values))

Accuracy : 0.8524590163934426
Precision : 0.8571428571428571
Recall : 0.8275862068965517
F1 : 0.8421052631578947

print(classification_report(y_pred, y_test))

```

	precision	recall	f1-score	support
0	0.85	0.88	0.86	32
1	0.86	0.83	0.84	29
accuracy			0.85	61
macro avg	0.85	0.85	0.85	61
weighted avg	0.85	0.85	0.85	61

Classification avec LinearSVC

```
hyperparametres = [
    {
        "penalty" : ["l1", "l2"],
        "loss" : ["squared_hinge"],
        "dual" : [False],
        "C" : [0.1, 1, 2, 3, 10],
    }
]

gscv_knn = GridSearchCV(
    estimator= LinearSVC(),
    param_grid= hyperparametres,
    cv = 5,
    scoring= "f1",
    verbose= 4
)

gscv_knn.fit(X_train, y_train.values.ravel())

Fitting 5 folds for each of 10 candidates, totalling 50 fits
[CV 1/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l1;,
score=0.714 total time= 0.0s
[CV 2/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l1;,
score=0.864 total time= 0.0s
[CV 3/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l1;,
score=0.829 total time= 0.0s
[CV 4/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l1;,
score=0.884 total time= 0.0s
[CV 5/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l1;,
score=0.727 total time= 0.0s
[CV 1/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l2;,
score=0.683 total time= 0.0s
[CV 2/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l2;,
score=0.837 total time= 0.0s
[CV 3/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l2;,
score=0.780 total time= 0.0s
[CV 4/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l2;,
score=0.837 total time= 0.0s
[CV 5/5] END C=0.1, dual=False, loss=squared_hinge, penalty=l2;,
score=0.744 total time= 0.0s
[CV 1/5] END C=1, dual=False, loss=squared_hinge, penalty=l1;,
score=0.718 total time= 0.0s
[CV 2/5] END C=1, dual=False, loss=squared_hinge, penalty=l1;,
score=0.837 total time= 0.0s
[CV 3/5] END C=1, dual=False, loss=squared_hinge, penalty=l1;,
score=0.800 total time= 0.0s
[CV 4/5] END C=1, dual=False, loss=squared_hinge, penalty=l1;,
score=0.884 total time= 0.0s
```

```
score=0.837 total time= 0.0s
[CV 5/5] END C=1, dual=False, loss=squared_hinge, penalty=l1,,
score=0.762 total time= 0.0s
[CV 1/5] END C=1, dual=False, loss=squared_hinge, penalty=l2,,
score=0.718 total time= 0.0s
[CV 2/5] END C=1, dual=False, loss=squared_hinge, penalty=l2,,
score=0.837 total time= 0.0s
[CV 3/5] END C=1, dual=False, loss=squared_hinge, penalty=l2,,
score=0.821 total time= 0.0s
[CV 4/5] END C=1, dual=False, loss=squared_hinge, penalty=l2,,
score=0.837 total time= 0.0s
[CV 5/5] END C=1, dual=False, loss=squared_hinge, penalty=l2,,
score=0.762 total time= 0.0s
[CV 1/5] END C=2, dual=False, loss=squared_hinge, penalty=l1,,
score=0.718 total time= 0.0s
[CV 2/5] END C=2, dual=False, loss=squared_hinge, penalty=l1,,
score=0.837 total time= 0.0s
[CV 3/5] END C=2, dual=False, loss=squared_hinge, penalty=l1,,
score=0.800 total time= 0.0s
[CV 4/5] END C=2, dual=False, loss=squared_hinge, penalty=l1,,
score=0.837 total time= 0.0s
[CV 5/5] END C=2, dual=False, loss=squared_hinge, penalty=l1,,
score=0.762 total time= 0.0s
[CV 1/5] END C=2, dual=False, loss=squared_hinge, penalty=l2,,
score=0.718 total time= 0.0s
[CV 2/5] END C=2, dual=False, loss=squared_hinge, penalty=l2,,
score=0.837 total time= 0.0s
[CV 3/5] END C=2, dual=False, loss=squared_hinge, penalty=l2,,
score=0.821 total time= 0.0s
[CV 4/5] END C=2, dual=False, loss=squared_hinge, penalty=l2,,
score=0.837 total time= 0.0s
[CV 5/5] END C=2, dual=False, loss=squared_hinge, penalty=l2,,
score=0.762 total time= 0.0s
[CV 1/5] END C=3, dual=False, loss=squared_hinge, penalty=l1,,
score=0.718 total time= 0.0s
[CV 2/5] END C=3, dual=False, loss=squared_hinge, penalty=l1,,
score=0.837 total time= 0.0s
[CV 3/5] END C=3, dual=False, loss=squared_hinge, penalty=l1,,
score=0.821 total time= 0.0s
[CV 4/5] END C=3, dual=False, loss=squared_hinge, penalty=l1,,
score=0.837 total time= 0.0s
[CV 5/5] END C=3, dual=False, loss=squared_hinge, penalty=l1,,
score=0.762 total time= 0.0s
[CV 1/5] END C=3, dual=False, loss=squared_hinge, penalty=l2,,
score=0.737 total time= 0.0s
[CV 2/5] END C=3, dual=False, loss=squared_hinge, penalty=l2,,
score=0.837 total time= 0.0s
[CV 3/5] END C=3, dual=False, loss=squared_hinge, penalty=l2,,
score=0.821 total time= 0.0s
```

```
[CV 4/5] END C=3, dual=False, loss=squared_hinge, penalty=l2;,
score=0.837 total time= 0.0s
[CV 5/5] END C=3, dual=False, loss=squared_hinge, penalty=l2;,
score=0.762 total time= 0.0s
[CV 1/5] END C=10, dual=False, loss=squared_hinge, penalty=l1;,
score=0.737 total time= 0.0s
[CV 2/5] END C=10, dual=False, loss=squared_hinge, penalty=l1;,
score=0.837 total time= 0.0s
[CV 3/5] END C=10, dual=False, loss=squared_hinge, penalty=l1;,
score=0.821 total time= 0.0s
[CV 4/5] END C=10, dual=False, loss=squared_hinge, penalty=l1;,
score=0.837 total time= 0.0s
[CV 5/5] END C=10, dual=False, loss=squared_hinge, penalty=l1;,
score=0.762 total time= 0.0s
[CV 1/5] END C=10, dual=False, loss=squared_hinge, penalty=l2;,
score=0.737 total time= 0.0s
[CV 2/5] END C=10, dual=False, loss=squared_hinge, penalty=l2;,
score=0.837 total time= 0.0s
[CV 3/5] END C=10, dual=False, loss=squared_hinge, penalty=l2;,
score=0.800 total time= 0.0s
[CV 4/5] END C=10, dual=False, loss=squared_hinge, penalty=l2;,
score=0.837 total time= 0.0s
[CV 5/5] END C=10, dual=False, loss=squared_hinge, penalty=l2;,
score=0.762 total time= 0.0s
```

```
GridSearchCV(cv=5, estimator=LinearSVC(),
              param_grid=[{'C': [0.1, 1, 2, 3, 10], 'dual': [False],
                           'loss': ['squared_hinge'], 'penalty': ['l1',
                           'l2']}],
              scoring='f1', verbose=4)
```